

VINEET BOTHRA

Delhi, India | vineetbothra1443@gmail.com | +91-7838344009

GitHub: github.com/Vins1443 | Portfolio: vineetbothra.com

EDUCATION

George Mason University, Fairfax, Virginia

Master's in Data Analytics and Engineering | **GPA: 3.70** | Aug 2022 - May 2024

IP University, Delhi, India

B.Tech in Information Technology | **GPA: 3.00** | May 2013 - May 2017

PROFESSIONAL EXPERIENCE

Sentry - United States

Senior Software Engineer | June 2023 - Feb 2025

- Extensive experience developing Django applications by building robust and scalable web solutions using Django's MVT (Model-View-Template) architecture. Expertise in creating reusable Views, Templates, and models, while optimizing CRUD operations using Django ORM.
- Developed and fine-tuned complex Django ORM queries to pre-load data, improving database performance and minimizing query overhead, which is critical for high-traffic applications.
- Created user-friendly, responsive UIs by integrating Django's templating language with modern frontend technologies like HTML5, CSS3, and JavaScript, ensuring seamless interaction between the backend and frontend.
- Designed and implemented Python APIs for various business-critical systems, including custom solutions to dump array structures for debugging processor failures, helping in error identification and resolution.
- Experience building RESTful APIs using both Django REST Framework and Flask, allowing for scalable communication between client-side applications and the server.
- Integrated Python solutions with AWS services like S3 and RDS, allowing for cloud-based storage and database management, streamlining static file hosting, and scalable database access.
- Designed and deployed Generative AI models using LangChain, OpenAI, and AWS Bedrock, improving automation capabilities by 40%.

Broadridge (Client) - Java R&D Pvt. Ltd., India

Python Developer | Jan 2021 - Aug 2022

- Implemented unit testing and automated test suites with Python and Java, ensuring robust CI/CD integration.
- Optimized Django ORM queries to pre-load high-volume data, minimizing query overhead and enhancing high-traffic performance.
- Participated in daily in-person meetings and adhere to Sprint, Scrum, and Agile development.
- Developed and executed RESTful APIs for online services and mobile applications by employing the Python Django-REST framework and the Flask Framework.
- Developed Python views, APIs, and database models to construct a responsive web application.
- Utilized Jenkins and GitHub to automate the construction and deployment of Docker containers in Mesos for Python source code version control.
- Applied Material-UI and Tailwind CSS for consistent, modern, and responsive UI design.
- Utilized Panda libraries to implement CSV file load procedures in Oracle using PySpark.
- Utilization of CRUD operations on Oracle SQL Database for data manipulation and retrieval.
- Complex database objects, such as stored procedures, functions, packages, and triggers, were generated through the use of SQL and PL/SQL.

Accenture - Gurugram, India

Associate Software Engineer (Python Developer) | Nov 2017 - Dec 2020

- Developed and automated Python scripts in AMAP, boosting project efficiency and client satisfaction by 40%.
- Enhanced database efficiency for Walmart with advanced schema normalization, boosting data retrieval speed by 20.3%.
- Led API development using AWS Lambda, API Gateway, and S3, reducing API latency and improving user experience.
- Experience with Pandas DataFrame operations, including filtering, grouping, and aggregating large datasets efficiently.
- Implemented React Router for handling navigation and dynamic routing within a single-page application (SPA).
- Developed AI-driven cohort analysis for Quinstreet using Tableau & Python, leading to a 14.6% rise in customer retention.
- Integrated Generative AI models into client solutions, improving decision-making efficiency by 35%.
- Enhanced Power BI dashboards for AMAP, optimizing performance and providing real-time business insights.
- Led Signet project's JavaScript, React, and Node.js development, driving 50,000+ monthly visitors.
- Deployed AWS solutions (EC2, S3, RDS), improving cloud scalability and client project delivery.

PROJECTS & CAPSTONE EXPERIENCE

AI-Powered Industry Code Prediction Model (Capstone, GMU)

- Led a **6-member team** to build an AI-driven **industry code classification model** using **LLMs and FDA technologies**.
- Optimized data processing with **Python, AWS, SQL, and Tableau**, reducing data loss and cutting processing time by **30%**.
- Developed ETL pipelines and real-time Kafka streams, improving data ingestion efficiency.

Generative AI Chatbot for Predictive Maintenance

- Engineered an **LLM-based chatbot** integrated with **LangChain** and **AWS Bedrock**, automating **machinery diagnostics and scheduling maintenance**.
- Utilized **Hugging Face transformers** for fine-tuning domain-specific chat responses.
- Improved incident resolution time by **50%** through AI-driven issue triage and root cause analysis.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Java, JavaScript, Typescript, C#, PHP, NoSQL
- **Front-End:** React Js, HTML, CSS, Material UI, Angular
- **Generative AI & NLP:** LangChain, OpenAI, Hugging Face, TensorFlow, AWS Bedrock.
- **Big Data & Data Engineering:** Kafka, Databricks, Apache Spark, ETL, AWS Glue, RDS.
- **Data Science & Analytics:** Supervised/Unsupervised Learning, Predictive Analytics, Feature Engineering.
- **BI & Visualization:** Tableau, Power BI, Advanced Excel.
- **Software Development & Backend:** Django, Flask, API Development, Unit Testing, Agile, Git.

- **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Jenkins, GitLab CI/CD, Azure.
 - **Other Tools:** Informatica, Jira, Databricks.
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CERTIFICATIONS & ACHIEVEMENTS

- **AWS Developer Associate Certification (August 2024)** – *Validation No: 5ae418756ef5455490ad5525e3c38443.*
- 10+ Certifications in **Data Science, AI/ML, Generative AI, and Data Engineering.**
- Top **5 position** in **Big Data Exploration Project** at GMU (*AIT 580 coursework – A grade*).